Simplex method and sensitivity analysis

# 1 Introduction

The simplex method is the first, and still probably the most popular, method to solve a linear program. While in theory we cannot guarantee it will converge to the optimal solution efficiently, in practice it often does. The algorithm is based on the fundamental concept of basic feasible solution, and it will also allow us to find a solution to the dual of the input problem. We will also discuss sensitivity analysis using the notations and the concepts of basis and basic feasible solutions. This allows us to be a bit more general and to reach more conclusions than in the discussion seen in class that avoids these concepts. Note that the material from these notes has not been covered in class, and will not be tested.

# 2 Basic Feasible Solutions

Let $c \in \mathbb{Z}^n$, $A \in \mathbb{Z}^{m \times n}$, $b \in \mathbb{Z}^m$. Throughout this lecture, we will consider LPs of the following form

$$\begin{array}{rlll} \max & c^T x & & \\ & Ax & = & b \\ & x & \geq & 0 \end{array} \tag{1}$$

where $A$ has full row rank. We say that such an LP is in *standard form*.

$\bar{x} \in \mathbb{R}^n$ is a *basic feasible solution* (bfs) for (1) if:

- $\bar{x}$ is feasible;
- There exists $m$ columns $B$ of $A$, called *basis*, such that, if we let
  - $A_B$ be the submatrix of $A$ corresponding to columns in $B$,
  - $N$ be the remaining columns,

  we have[1]

$$\bar{x} = \begin{pmatrix} \bar{x}_B \\ \bar{x}_N \end{pmatrix} = \begin{pmatrix} A_B^{-1} b \\ 0 \end{pmatrix}.$$

We are also going to assume in the following that the LP (1) is *non-degenerate*[2]. The following fact explains the relevance of the concept of bfs.

**Theorem 1.** *If an LP in standard form has an optimal solution, then it has one that is a basic feasible solution.*

[1] In particular, $A_B$ needs to be invertible.

[2] (1) is non-degenerate if $\bar{x}_B$ has no component equal to 0. This implies that there are no two bases that correspond to the same basic feasible solution.

Note moreover that each bfs is (uniquely, since the problem is non-degenerate) described by its *basis*, i.e., the $m$ columns in $B$. The next result shows that, from an optimal bfs, we can also obtain an optimal solution for the dual of (1), which following the classical conversion table can be described as follows:

$$\begin{array}{lll} \min & b^T y & \\ & A^T y \geq & c \\ & y & \text{free} \end{array} \tag{2}$$

**Theorem 2.** *Let $\bar{x}$ be a bfs corresponding to a basis $B$. $\bar{x}$ is optimal for* (1) *if and only if $\bar{y} = (c_B^T A_B^{-1})^T$ is an optimal solution to the dual of* (2). *The latter happens if and only if $\bar{y}$ is feasible for* (2).

*Proof.* Let $B$ be a basis with bfs $\bar{x}$ associated to it. Notice that we are not assuming optimality of $\bar{x}$ for the moment. Let $\bar{y} = (c_B^T A_B^{-1})^T$. Then

$$c^T \bar{x} = c_B^T \bar{x}_B = c_B^T A_B^{-1} b = \bar{y}^T b.$$

Thus, $\bar{x}$ and $\bar{y}$ always have the same objective function value. By strong duality, $\bar{y}$ is optimal for (2) if and only if it is feasible for (2). This shows that, if $\bar{y}$ is feasible for the dual, then it is optimal and $\bar{x}$ is optimal for the primal.

Now assume that $\bar{y}$ is not feasible for the dual. Then clearly $\bar{y}$ is not optimal for the dual. We claim that $\bar{x}$ is not optimal for the primal, concluding the proof. Since $\bar{y}$ is not feasible for the dual, there exists $i \in \{1, \ldots, n\}$ such that $a^{i,T}\bar{y} < c_i$, where $a^i$ is the $i$-th column of matrix $A$ and $a^{i,T}$ is its transpose. First we claim that $i \notin B$. Indeed, for $i \in B$, we have

$$a^{i,T}\bar{y} = \bar{y}^T a^i = c_B^T A_B^{-1} a^i = c_i.$$

Thus, $i \notin B$. Let $\epsilon > 0$ be small enough and consider the point $\tilde{x}$ defined as follows:

$$\tilde{x}_j = \begin{cases} \bar{x}_j - (A_B^{-1} a^i) \cdot \epsilon & \text{for } j \in B \\ \epsilon & \text{for } j = i \\ 0 & \text{otherwise} \end{cases}$$

We claim that $\tilde{x}$ is feasible for (1) and $c^T \tilde{x} > c^T \bar{x}$, showing that $\bar{x}$ is not optimal. To show feasibility, observe that $\tilde{x} \geq 0$ by choice of $\epsilon$ and non-degeneracy. Moreover,

$$A\tilde{x} = A_B \tilde{x}_B + A_N \tilde{x}_N = \underbrace{A_B \bar{x}_B}_{b} - A_B (A_B^{-1} a^i) \cdot \epsilon + \epsilon \cdot a^i = b - \epsilon \cdot a^i + \epsilon \cdot a^i = b.$$

This shows that $\tilde{x}$ is feasible. We can thus conclude the proof by observing

$$c^T \tilde{x} = c_B^T \tilde{x}_B + c_N^T \tilde{x}_N = c_B^T \bar{x}_B - \underbrace{c_B^T A_B^{-1}}_{\bar{y}^T} a^i \cdot \epsilon + c_i \epsilon = c^T \bar{x} + \epsilon \underbrace{(c_i - \bar{y}^T a^i)}_{>0} > c^T \bar{x}.$$

□

# 3 The simplex method

The previous facts on bfs suggest an algorithm for finding the optimal solution of an LP in standard form[3]. This algorithm goes under the name of simplex method. In the following, we call a basis $\tilde{B}$ *adjacent* to $B$ if it only differs by a column.

[3] Note that the algorithm has to be modified a bit to be able to detect than an LP is unbounded, and to find a bfs to start from. This is however not hard to do.

1. Let $B$ be a basis and $\bar{x}$ be the corresponding bfs;
2. Construct the dual solution $\bar{y} = (c_B^T A_B^{-1})^T$;
   (a) If $\bar{x}, \bar{y}$ are optimal: **stop** we have found the optimal solution;
   (b) Else:
      - Let $\tilde{B}$ be a basis adjacent to $B$ such that the corresponding bfs $\tilde{x}$ satisfies $c^T \tilde{x} > c^T \bar{x}$;
      - Change $B$ to $\tilde{B}$, $\bar{x}$ to $\tilde{x}$, and go to 2.

Note that the algorithm does no specify how to find an adjacent basis to the current one. One can of course try all possibilities, but there are faster ways to do so based on the solution of linear systems of equations. Since at every step of the algorithm we strictly increase the objective function vale, the algorithm converges in finite time with a solution, which can be proved to be optimal.

**Theorem 3.** *The simplex method is well-defined and terminates after a finite number of steps to an optimal solution of the LP.*

# 4 Sensitivity analysis

Suppose we have solved the LP (1) and obtained an optimal bfs $\bar{x}$ corresponding to a basis $B$. Let $\bar{y}$ be the corresponding optimal dual solution. In this section, we investigate how the optimal primal solution changes when one of the following happens:

- A new variable is added;
- A new constraint is added;
- An objective function coefficient changes;
- The right-hand side of one constraint changes;

Such scenarios arise when data (eg, market conditions, production costs,...) changes, or when we would like to understand the what happens should it change to assess the robustness of our solution. Let us consider each of the above situations separately.

## 4.1 Addition of a variable

Suppose a new variable $x_0$ is added. Note that $B$ is still a basis and in the new problem it corresponds to the bfs $\tilde{x} = (\bar{x}, 0)$, obtained from $\bar{x}$ by adding a 0 to the component corresponding to the new variable. However, $\tilde{x}$ may not be optimal for the original problem - we can check it by computing the dual solution $\tilde{y}$ corresponding to the basis $B$ in the new problem, and check if $\tilde{y}$ is feasible. If $\tilde{x}$ is not optimal, one can solve the new LP starting the simplex method with the basis $B$. This is often (much) faster that solving the new primal problem from scratch.

## 4.2 Addition of a constraint

Suppose now a new constraint $a^T x = \beta$ is added to (1). Note that, if $\bar{x}$ is feasible for the new LP, it is also optimal, since the new feasible region is contained in the previous one. However, $\bar{x}$ may not satisfy the new constraint, and thus be infeasible. In which case, we cannot use $\bar{x}$ as a starting solution for the simple method.

But not all is lost. Indeed, adding a constraint to the primal corresponds to additing a variable to the dual. Following the argument from Section 4.1, the solution $\tilde{y} = (\bar{y}, 0)$ is feasible for the new dual (where again the new component with value 0 corresponds to the new variable). We can then solve the dual starting from $\tilde{y}$ and, once the optimum is obtained, deduce an optimal solution for the primal from the optimal dual basis. Such a procedure is called *dual simplex* method.

## 4.3 Change of one coefficient in the objective function

Suppose now that coefficient of one variable in the objective function changes. Without loss of generality, assume that this variable is $x_1$, and its objective function coefficient changes from $c_1$ to $c_1' = c_1 + \Delta$. Notice that $\bar{x}$ stays a bfs associated to the same basis $B$. However, it may not be optimal anymore. In the following, we compute the interval of $\Delta$ that keeps $\bar{x}$ optimal. Let $a^1$ be the first column of matrix $A$ and $a^{1,T}$ its transpose. Also, let $[A_B^{-1}]_1$ be the first row of $A_B^{-1}$.

**Theorem 4.**

- *If $1 \in B$ (i.e., column 1 is in the optimal basis of the original problem), then $\bar{x}$ is optimal for the new problem if and only if $c_B^T A_B^{-1} a^i + \Delta [A_B^{-1}]_1 a^i - c_i \geq 0$ for every non-basic variable $i$.*
- *Else, $1 \notin B$ and $\bar{x}$ is optimal for the new problem if and only if $\Delta \leq c_B^T (A_B^{-1} a^1) - c_1$.*

*Proof.* By Theorem 2, $\bar{x}$ is optimal for the new primal if and only if the new corresponding dual solution is feasible for the new dual. Recall that $\bar{y} = (c_B^T A_B^{-1})$ and that the feasibility constraints are of the form $A^T y \geq c$.

Suppose first that column 1 is not in the basis $B$. Then the new dual solution does not change (since neither $c_B$, nor $A_B$ change). Thus, $\bar{y}$ is optimal if and only if it is feasible for the new dual. Since the only dual constraints that has changed is the one corresponding to $x_1$, it is only one we need to check. In order to satisfy the new constraint, we must have

$$a^{1,T} \bar{y} \geq c_1 + \Delta \;\Leftrightarrow\; \Delta \leq a^{1,T} (A_B^{-1})^T c_B - c_1 = c_B^T (A_B^{-1} a^1) - c_1,$$

as required.

Now assume $1 \in B$. Then the new dual solution $\tilde{y}$ associated to $B$ has value

$$\tilde{y}^T = c_B^T A_B^{-1} + \Delta [A_B^{-1}]_1,$$

and again $\bar{x}$ is optimal if and only if $\tilde{y}$ is feasible for the dual. Notice however that we now have to check all constraints of the dual (since the dual solution has changed). For the first constraint, we have

$$a^{1,T} \tilde{y} \geq c_1 + \Delta \;\Leftrightarrow\; \tilde{y}^T a^1 - c_1 - \Delta \geq 0 \Leftrightarrow (c_B^T A_B^{-1} + \Delta [A_B^{-1}]_1) a^1 - c_1 - \Delta \geq 0 \;\Leftrightarrow\; c_1 + \Delta - c_1 - \Delta \geq 0,$$

which is trivially satisfied. For every other constraint $i \in B$, we must have

$$a^{i,T} \tilde{y} \geq c_i \;\Leftrightarrow\; \tilde{y}^T a^i - c_i \geq 0 \Leftrightarrow (c_B^T A_B^{-1} + \Delta [A_B^{-1}]_1) a^i - c_i \geq 0 \;\Leftrightarrow\; c_i + 0 - c_i \geq 0,$$

which is also trivially satisfied. Thus, the only conditions left are those corresponding to a non-basic variable $i$, for which we must have

$$a^{i,T} \tilde{y} \geq c_i \;\Leftrightarrow\; \tilde{y}^T a^i - c_i \geq 0 \Leftrightarrow c_B^T A_B^{-1} a^i + \Delta [A_B^{-1}]_1 a^i - c_i \geq 0,$$

as required. □

## 4.4 Change of the right-hand side

Suppose now that the right-hand side of the a primal constraint – say, without loss of generality, the first – changes from $b_1$ to $b_1 + \Delta$ for some $\Delta \neq 0$. Clearly, $\bar{x}$ is not feasible anymore (since it satisfies $a^{1,T}\bar{x} = b_1 \neq b_1 + \Delta$). It may however be that the new basic solution $\tilde{x}$ associated to $B$ is feasible, and even optimal. The following theorem gives condition under which this holds.

**Theorem 5.** *$\tilde{x}$ is optimal if and only if $A_B^{-1}b + [A_B^{-1}]^1\Delta \geq 0$, where $[A_B^{-1}]^1$ is the first column of $A_B^{-1}$. Moreover, when going from $\bar{x}$ to $\tilde{x}$, the objective function value changes by $\Delta\bar{y}_1$.*

*Proof.* First observe that

$$\tilde{x}_B = A_B^{-1}b + [A_B^{-1}]^1\Delta \qquad \text{and} \qquad \tilde{x}_N = 0.$$

For feasibility of $\tilde{x}$ in the new LP, it is enough to have

$$\tilde{x} \geq 0 \ \Leftrightarrow \ A_B^{-1}b + [A_B^{-1}]^1\Delta \geq 0.$$

Now let us consider optimality of $\tilde{x}$. Observe that, $\bar{y}^T = c_B A_B^{-1}$, so $\bar{y}$ is the dual solution corresponding to basis $B$ for the new problem. Moreover, since the dual constraints have not changed, $\bar{y}$ is feasible for the new dual. Thus, $\tilde{x}$ is optimal for the new primal if and only if it is feasible for the new primal. This shows the first part of the statement.

For the second part, observe

$$c^T\tilde{x} = c_B^T\tilde{x}_B = c_B^T\bar{x}_B + c_B^T[A_B^{-1}]^1\Delta = c^T\bar{x} + \Delta\bar{y}_1.$$

□

The second statement in the theorem above is the reason why the dual solution $\bar{y}$ is called the vector of *shadow prices*: $\bar{y}_1$ tells us how much the objective function changes when the right-hand side of the first constraint increases by 1. In a production model like the table/chairs model seen multiple times in class, the first constraint is imposed by the availability of a certain resource (e.g., wood). Thus, $\bar{y}_1$ denotes how much we are willing to spend to buy one unit of wood (since having 1 more unit of wood, i.e. $\Delta = 1$, will let us increase the objective function value by $\bar{y}_1$).